AI-HOS | Hospital Operating System

Complete System Workflow & Architecture Document

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52	103	18	8
Pages	Components	API Endpoints	User Roles

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1. System Architecture

Tech Stack

Layer	Technology
Frontend	Next.js 15, React 19, TypeScript, Tailwind CSS, shadcn/ui
Backend	Next.js API Routes, Server Actions
Database	Supabase (PostgreSQL) with Row Level Security
Authentication	NextAuth.js v5 (JWT strategy, 12h sessions)
Queue System	BullMQ + Redis (separate worker container)
Caching	Redis (rate limiting, sessions)
WhatsApp	Meta Cloud API (direct integration)
AI	Claude API (WhatsApp AI agent)
Deployment	Docker + Traefik + Cloudflare
PWA	Service Worker, Web Manifest, VAPID Push

Multi-Tenancy Hierarchy

Platform (Super Admin) → **Client (Hospital Group)** →
Branch (Individual Hospital) → **Users (Doctors, Staff, Patients)**

Every database row is isolated by `tenant_id` using Supabase RLS policies. A Client Admin manages multiple branches. Super Admin manages all clients.

Subscription Tiers

Feature	Basic	Medium	Enterprise
OPD Queue & Appointments	Yes	Yes	Yes
Pharmacy Module	-	Yes	Yes
Lab Module	-	Yes	Yes
IPD (Inpatient)	-	-	Yes
Telemedicine (Video)	-	Yes	Yes

ABDM / ABHA	-	-	Yes
WhatsApp AI Agent	-	Yes	Yes
Advanced Analytics	-	-	Yes

2. User Roles & Access Control

Role	Login Method	Access Areas	Key Capabilities
SUPER_ADMIN	Email + PIN	/platform/*	Manage all clients, branches, subscriptions, platform settings, WhatsApp routing, system health
CLIENT_ADMIN	Client ID + PIN	/admin/*, /reception/*, /doctor/*, /pharmacy/*, /lab/*	Manage all branches under a client, switch branches, manage doctors/staff/patients
BRANCH_ADMIN	Tenant ID + PIN	/admin/*, /reception/*, /doctor/*, /pharmacy/*, /lab/*	Manage single branch: doctors, patients, staff, billing, settings, analytics
RECEPTION	Tenant ID + PIN	/reception/*	Queue board, walk-in booking, appointments, patient search, check-in, admissions
DOCTOR	Doctor ID + PIN	/doctor/*	Dashboard, consult patients, prescriptions, EMR, video consult, schedule
PHARMACIST	Staff ID + PIN	/pharmacy/*	Prescription orders (Kanban), inventory management
LAB_TECH	Staff ID + PIN	/lab/*	Lab order Kanban (Pending → In Progress → Completed), result entry
PATIENT	Phone + WhatsApp OTP	/patient/*	Dashboard, book appointments, view prescriptions, lab results, OP passes, invoices, profile

3. Authentication Workflows

Hospital Staff Login (PIN-based)

User selects role & branch → Enters PIN → Server validates PIN against tenant record → Rate limit: 5 attempts / 15 min (Redis) → JWT issued (12h expiry) → Custom Supabase JWT for RLS → Redirect to role dashboard

Hospital Staff Login (Password-based)

User enters email + password → Supabase Auth validates credentials → Lookup user_credentials table for role mapping → Verify tenant is active → JWT issued with role/tenant

Patient Login (OTP via WhatsApp)

Patient enters phone number → API checks patient exists (phone variants: 10-digit + 91-prefixed) → Rate limit: 3 OTPs / phone / 15 min + 10 OTPs / IP / 15 min → 6-digit OTP generated & stored in patient_otps table → OTP sent via WhatsApp Cloud API → Patient enters OTP (5 min validity) → Server verifies OTP (both phone formats matched) → JWT issued with PATIENT role → Redirect to /patient dashboard

4. Reception & Queue Management

Walk-in Patient Booking

Reception opens /reception/book → Search existing patient or create new → Select doctor & time slot → Booking created with token number → Patient added to queue (status: WAITING) → WhatsApp confirmation sent to patient → Token receipt printable

Queue Board (/reception)

- **Real-time stats banner**: Total waiting, in-consultation, completed, avg wait time
- **Queue cards**: Token #, patient name, doctor, wait time (elapsed timer)
- **Status flow**: WAITING → CHECKED_IN → IN_CONSULTATION → COMPLETED
- **Quick actions**: Check-in, Mark arrived, Cancel, Reschedule
- **Real-time updates**: Supabase realtime subscriptions

Appointment Management (/reception/appointments)

- Calendar view of all appointments
- Filter by doctor, date range, status
- Reschedule and cancel with WhatsApp notification
- Pending arrivals panel with quick check-in

Patient Search (/reception/patients)

- Search by name, phone, patient ID
- View patient history, past visits, documents
- Quick book appointment from patient record

5. Doctor Consultation Workflow

Doctor Dashboard (/doctor)

- Today's queue: patients waiting, in consultation
- Stats: total patients today, completed, pending
- Quick access to current patient

Consultation Flow (/doctor/consult)

Doctor opens consult page → Sees queue of checked-in patients → Clicks patient to start consultation → Records: Chief complaints, diagnosis, vitals, clinical notes → Writes prescription (medicine combobox with templates) → Orders lab tests (if needed) → Admits

to IPD (if needed) → Video call button (telemedicine, feature-gated) → Marks consultation complete → Post-consultation notification sent via WhatsApp → Prescription printable / PDF exportable

EMR - Electronic Medical Record (/doctor/emr/[phone])

- Complete patient history by phone number
- Past consultations, prescriptions, lab results, vitals timeline
- Document gallery (uploaded reports, images)

Prescription Management

- Medicine combobox with auto-complete
- Prescription templates (save & reuse common prescriptions)
- Print prescription (branded hospital letterhead)
- Download as PDF

Doctor Schedule (/doctor/schedule)

- Set weekly availability (day + time slots)
- Block specific dates
- Visible to patients during booking

6. Pharmacy Module

Feature-gated: Medium & Enterprise tiers

Prescription Orders (/pharmacy)

Doctor writes prescription → Order appears in Pharmacy Kanban → **Pharmacist** sees: Pending → Preparing → Ready → Dispensed → Pharmacist moves cards through columns → Patient notified when ready for pickup

Inventory Management (/pharmacy/inventory)

- Medicine stock tracking
- Low stock alerts
- Batch and expiry tracking
- Add/edit/delete medicines

7. Lab Module

Feature-gated: Medium & Enterprise tiers

Lab Order Workflow (/lab)

Doctor orders lab test during consultation → Order appears in Lab Kanban board → **Lab Tech** sees: Pending → Sample Collected → In Progress → Completed → Lab Tech enters results with normal range indicators → Results visible to Doctor in EMR → Results visible to Patient in portal → Lab report printable with normal ranges

8. Inpatient (IPD) Module

Feature-gated: Enterprise tier only

Admission Flow

Doctor/Reception initiates admission from consult → Admit dialog: select bed, ward, admission notes → Patient status changes to ADMITTED → Appears on Inpatient board (/reception/admissions)

IPD Components

Component	Purpose
Bed Map	Visual ward/bed occupancy view
Admission Detail	Full inpatient record
Vitals Chart	Temperature, BP, pulse over time
Nursing Notes	Shift-wise nursing documentation
Rounds Panel	Doctor rounds notes & orders
Daily Charges	Room charges, procedures, consumables
Transfer Dialog	Transfer between wards/beds
Discharge Dialog	Discharge summary + final billing

Discharge Flow

Doctor initiates discharge → Discharge summary (diagnosis, treatment, follow-up) → Final bill generated (all daily charges + procedures) → Discharge summary printable → Bed freed on bed map

9. Billing & Payments

Billing Workflow (/admin/billing)

Consultation completed → Invoice auto-generated (consultation fee + lab + pharmacy + procedures) → GST/tax calculation (lib/billing/tax.ts) → Payment: Cash / Razorpay (online) → Razorpay callback verifies payment → Invoice printable / PDF → Payment receipt sent via email

Invoice Print Components

- Invoice with hospital branding, GST, itemized charges
- OP consultation summary print
- Patient card print
- All printable via PrintButton or DownloadPdfButton

10. Patient Portal

Patient Dashboard (/patient)

- Upcoming appointments with countdown
- Recent prescriptions and lab results
- Quick book appointment button

Patient Features

Page	Functionality
/patient/book	Book new appointment: select doctor, date, time slot
/patient/appointments	View upcoming & past appointments, cancel/reschedule
/patient/prescriptions	View all prescriptions, print/download PDF
/patient/lab	View lab results with normal range indicators
/patient/op-pass	OP (outpatient) pass with QR verification
/patient/invoices	View and download invoices
/patient/profile	View/edit personal details, ABHA linkage

OP Pass Verification

Patient books appointment → OP Pass generated with unique ID → QR code on pass → Reception scans QR at /verify/[id] → Patient verified & checked in

11. Platform Administration (Super Admin)

Platform Dashboard (/platform)

- Global stats: total clients, branches, doctors, patients, revenue
- System health monitoring

Platform Management Pages

Page	Purpose
/platform/clients	Manage hospital groups (create, edit, activate/deactivate)
/platform/clients/[id]	Client detail: branches, subscription, settings
/platform/branches	All branches across all clients
/platform/doctors	All doctors across platform
/platform/patients	All patients across platform
/platform/plans	Subscription plan management (Basic/Medium/Enterprise)
/platform/analytics	Global analytics and trends
/platform/settings	Platform-wide configuration
/platform/logs	Audit log viewer
/platform/health	System health: DB, Redis, API status
/platform/whatsapp	WhatsApp routing: per-tenant phone number assignment

Client Onboarding Wizard

Super Admin clicks "New Client" → Onboarding wizard: Client name, contact, subscription tier → Create first branch (hospital name, address, PINs) → Configure features based on tier → Client Admin credentials generated → Client ready to use

12. WhatsApp AI Agent

Feature-gated: Medium & Enterprise tiers

Architecture

Patient sends WhatsApp message → Meta Webhook → /api/whatsapp/webhook → Message deduplication → Session management (conversation context) → Claude AI Agent processes message → Tool calls: book appointment, check status, get prescriptions, etc. → Response sent back via WhatsApp Cloud API

AI Agent Tools

Tool	What it does
book_appointment	Book appointment with doctor
check_appointment_status	Check upcoming appointments
cancel_appointment	Cancel an existing appointment
get_prescriptions	Retrieve recent prescriptions
get_lab_results	Retrieve lab test results
get_doctors	List available doctors & specialties
get_available_slots	Check doctor availability
get_queue_status	Check current queue position

Multi-language Support

The WhatsApp agent responds in the language the patient uses (English, Hindi, Telugu). Translation layer in lib/whatsapp/translations.ts.

13. ABDM / ABHA Integration

Feature-gated: Enterprise tier only

ABHA (Ayushman Bharat Health Account)

- **ABHA Card**: Display patient's ABHA ID and QR code
- **ABHA Link Dialog**: Link existing ABHA to patient record
- **Consent Manager**: Manage health data sharing consents
- **Health Record Export**: Export records in FHIR format

FHIR Integration (lib/abdm/fhir.ts)

Converts hospital records to FHIR R4 format for interoperability with national health systems.

14. Telemedicine

Feature-gated: Medium & Enterprise tiers

Video Consultation Flow

Doctor clicks "Video Call" during consultation → Jitsi Meet room created (unique per appointment) → Full-screen video dialog opens in browser → "Copy Link for Patient" copies Jitsi URL → Link sent to patient via WhatsApp API → Patient joins from their phone/computer → Doctor can consult, share screen → No API keys needed (Jitsi is free/open source)

15. Notifications & Reminders

WhatsApp Notifications (via Meta Cloud API)

Event	Notification
Appointment booked	Booking confirmation with doctor, date, time
Appointment cancelled	Cancellation confirmation
24h before appointment	Reminder notification
2h before appointment	Final reminder
Consultation completed	Post-consultation summary

Queue update	Your turn is approaching
OTP for login	6-digit verification code

Push Notifications (VAPID / Service Worker)

- Browser push notifications for PWA users
- Subscription stored in Supabase
- Triggered by BullMQ workers

BullMQ Background Workers

Worker	Purpose
reminder-worker	Send appointment reminders (24h and 2h before)
notification-worker	Process WhatsApp message queue
cleanup-worker	Clean expired OTPs, old sessions
report-worker	Generate daily/weekly summary reports

16. Infrastructure & Deployment

Docker Architecture

Container	Image	Purpose
ai-hos-ai-hos-1	Next.js standalone	Main app (web + API)
ai-hos-ai-hos-worker-1	Node.js Alpine	BullMQ workers (reminders, notifications)
ai-hos-bullboard-1	BullBoard	Queue monitoring dashboard
redis	Redis	Rate limiting, BullMQ, caching
traefik	Traefik	Reverse proxy, SSL termination

PWA Features

- Service Worker with stale-while-revalidate caching
- Installable on desktop, Android, iOS
- PNG icons (192px, 512px, Apple touch icon)
- Push notification support
- App shortcuts: Queue Board, Book Appointment

i18n (Internationalization)

- 3 languages: English, Hindi, Telugu
- 95 translation keys across all modules
- Language switcher in sidebar
- Persisted in localStorage

Security

- Redis rate limiting on all auth endpoints
- Supabase RLS for tenant data isolation
- Custom JWT for client-side Supabase access
- Security headers: HSTS, X-Frame-Options: DENY, CSP
- OTP anti-enumeration (generic error messages)
- PIN brute-force protection (5 attempts / 15 min)

17. API Reference

Endpoint	Method	Purpose

/api/auth/[...nextauth]	GET/POST	NextAuth.js authentication handler
/api/auth/tenants	GET	List available tenants for login
/api/auth/forgot-password	POST	Send password reset email
/api/auth/create-password	POST	Set new password
/api/patient-auth/send-otp	POST	Send OTP via WhatsApp
/api/booking	POST	Create staff-side booking
/api/patient-booking	POST	Create patient-side booking
/api/patient-appointment	GET/PATCH	Patient appointment management
/api/queue/notify	POST	Send queue position notification
/api/notifications/post-consultation	POST	Send post-consultation summary
/api/payment/callback	POST	Razorpay payment verification
/api/payment/send-emails	POST	Send payment receipt emails
/api/telemedicine/send-link	POST	Send video call link via WhatsApp
/api/push/subscribe	POST	Store push notification subscription
/api/whatsapp/webhook	GET/POST	WhatsApp webhook (verify + receive messages)
/api/abdm	POST	ABDM integration operations
/api/platform	GET/POST	Platform admin operations
/api/verify/[id]	GET	OP Pass QR verification

18. Database Schema Overview

Core Tables

Table	Purpose	Key Columns
clients	Hospital groups	client_id, name, admin_pin, status
tenants	Individual branches	tenant_id, client_id, hospital_name, admin_pin, reception_pin, status
tenant_features	Feature flags per branch	tenant_id, feature_key, enabled
doctors	Doctor records	doctor_id, tenant_id, name, specialty, pin
staff	Non-doctor staff	staff_id, tenant_id, name, role, pin
patients	Patient records	phone, name, tenant_id, dob, gender
platform_admins	Super admin accounts	admin_id, email, name, pin
user_credentials	Email/password login mapping	email, role, entity_table, entity_id, tenant_id

Clinical Tables

Table	Purpose
appointments / bookings	Appointment records with status tracking
consultations	Consultation notes, diagnosis, vitals
prescriptions	Prescription records with medicines
lab_orders	Lab test orders with results
pharmacy_orders	Pharmacy dispensing orders
admissions	IPD admission records
invoices	Billing and payment records

System Tables

Table	Purpose
patient_otps	OTP storage for patient login
audit_logs	System-wide audit trail

push_subscriptions	Web push notification subscriptions
whatsapp_sessions	AI agent conversation context
whatsapp_routes	Per-tenant WhatsApp phone number routing

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Production URL: https://app.ainewworld.in
VPS: 72.61.238.6 | Docker containers: 3 (app + worker + bullboard)
All data isolated by tenant_id with Supabase Row Level Security.